



Semiring isomorphisms between rational function semifields of tropical curves



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ABSTRACT

We prove that a semiring isomorphism between the rational function semifields of two tropical curves induces an expansive map between those tropical curves. This semiring isomorphism and the expansive map respect zeros and poles of rational functions with their degrees. As a corollary, we show that the automorphism group of a tropical curve is isomorphic to the T -algebra automorphism group of its rational function semifield, where $T := (\mathbf{R} \cup \{-\infty\}, \max, +)$ is the tropical semifield. Finally, we describe all semiring automorphisms of rational function semifields of all tropical curves.

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1. Introduction

By an operation called tropicalization, n -dimensional varieties correspond to n -dimensional polyhedral complexes called tropical varieties. The relation between varieties and tropical varieties is one of the biggest interests to study in tropical geometry; e.g., [10] is a great article studying amoebae, which are intermediate objects between varieties and tropical varieties, and contains an excellent survey on them.

The fundamental theorem of tropical geometry [19, Theorem 3.2.3] (Kapranov's Theorem in the case of a hypersurface), which provides the geometric meaning of tropical varieties, gives us the today's foundation of abstract tropical geometry; abstract tropical varieties are defined and studied by many authors (e.g., [1], [2], [4], [6], [18], [20], [21]). To establish theories of abstract tropical varieties, we often use facts which hold for varieties in the classical algebraic geometry as working hypotheses. In this paper, we focus on rational

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function semifields of tropical curves to shed light on intrinsic structure of abstract tropical geometry and consider the following well-known fact:

The category of nonsingular projective curves and dominant morphisms is equivalent to the category of function fields of dimension one over k and k -homomorphisms, where k is an algebraically closed field (see [8, Corollary. 6.12. in Chapter I]).

Is there a tropical analogue of this fact? We factorize this question into the following three subquestions.

(1) Is the rational function semifield $\text{Rat}(\Gamma)$ of a tropical curve Γ a finitely generated semifield over the tropical semifield $\mathbf{T} := (\mathbf{R} \cup \{-\infty\}, \max, +)$? If so, does $\text{Rat}(\Gamma)$ have dimension one over $\mathbf{T}$ (with some proper definition of dimension)?

(2) What kind of semifields over $\mathbf{T}$ determines a tropical curve? In particular, if (1) holds, does a finitely generated semifield over $\mathbf{T}$ of dimension one determine a tropical curve? How do we construct it?

(3) For two tropical curves Γ and Γ' , does a $\mathbf{T}$ -algebra homomorphism $\text{Rat}(\Gamma) \rightarrow \text{Rat}(\Gamma')$ induce a finite harmonic morphism $\Gamma' \rightarrow \Gamma$?

In [12], the author considered question (1). The main theorem of [12] states that $\text{Rat}(\Gamma)$ is a finitely generated semifield over $\mathbf{T}$. The last part of question (2) is partially solved by [14, Proposition 3.12]. The rests of (1) and (2) are still open. But the work in [15] is related to question (2). In [15], Jun deals with three kinds of valuations of semirings, *classical*, *strict* and *hyperfield valuations*, and has constructed the abstract curve associated with the rational function semifield over $(\mathbf{Q} \cup \{-\infty\}, \max, +)$ via strict valuations. Note that in the classical case, the Krull dimension of the coordinate ring of a variety coincides with the transcendence degree of its quotient field. However, in the tropical case, there is no definition of transcendence degree of semifields. For the definition of Krull dimension of semirings, see [17]. Question (3) is solved by [13, Theorem 1.1] and [14, Corollary 3.17] with different ways.

A special case of question (3) is our main topic:

(3)' For two tropical curves Γ and Γ' , does a $\mathbf{T}$ -algebra isomorphism $\text{Rat}(\Gamma) \rightarrow \text{Rat}(\Gamma')$ induce an isomorphism (i.e., a finite harmonic morphism of degree one) $\Gamma' \rightarrow \Gamma$?

The answer is YES. The following is our main theorem:

Theorem 1.1. *Let Γ, Γ' be tropical curves. Let $\psi : \text{Rat}(\Gamma) \rightarrow \text{Rat}(\Gamma')$ be a semiring isomorphism between their rational function semifields. Then $\psi(1) \in \mathbf{R}_{>0}$ and ψ induces a $\frac{1}{\psi(1)}$ -expansive map $\varphi : \Gamma' \rightarrow \Gamma$. In particular, when ψ is a $\mathbf{T}$ -algebra isomorphism, $\psi(1) = 1$ and φ is a finite harmonic morphism of degree one.*

Here, for $r \in \mathbf{R}_{>0}$, an r -expansive map between tropical curves $\varphi' : \Gamma \rightarrow \Gamma'$ means a continuous bijection which induces a metric isomorphism between $\Gamma \setminus \Gamma_\infty$ and $\Gamma' \setminus \Gamma'_\infty$ with r as its expansion factor, i.e., $r \cdot \text{dist}(x, y) = \text{dist}(\varphi'(x), \varphi'(y))$ for any $x, y \in \Gamma \setminus \Gamma_\infty$, where $\text{dist}(x, y)$ denotes the distance between x and y and $\cdot$ denotes the usual multiplication of real numbers and we define $r \cdot \infty = \infty$.

Note that [13, Theorem 1.1] and [14, Corollary 3.17] give also an affirmative answer for question (3)', but they are only in $\mathbf{T}$ -algebra setting, and Theorem 1.1 is considered in semiring setting.

One of the most important consequences of Theorem 1.1 is that the $\mathbf{T}$ -algebra structure of the rational function semifield of a tropical curve perfectly determines (not only the topological structure but) the metric structure of the tropical curve.

Theorem 1.1 has the following corollary:

Corollary 1.2. *The following groupoids $\mathcal{C}, \mathcal{D}$ are isomorphic.*

(1) *The class $\text{Ob}(\mathcal{C})$ of objects of $\mathcal{C}$ is the tropical curves.*

For $\Gamma, \Gamma' \in \text{Ob}(\mathcal{C})$, the set $\text{Hom}_{\mathcal{C}}(\Gamma, \Gamma')$ of morphisms from Γ to Γ' consists of the semiring isomorphisms $\text{Rat}(\Gamma) \rightarrow \text{Rat}(\Gamma')$.

(2) *The class $\text{Ob}(\mathcal{D})$ of objects of $\mathcal{D}$ is the tropical curves.*

For $\Gamma, \Gamma' \in \text{Ob}(\mathcal{D})$, the set $\text{Hom}_{\mathcal{D}}(\Gamma, \Gamma')$ of morphisms from Γ to Γ' consists of the r -expansive maps $\Gamma \rightarrow \Gamma'$ for all $r \in \mathbf{R}_{>0}$.

In particular, considering the case that $\Gamma = \Gamma'$ and ψ is a $\mathbf{T}$ -algebra automorphism of $\text{Rat}(\Gamma)$, we have the following corollary:

Corollary 1.3. *The automorphism group of a tropical curve Γ is isomorphic to the $\mathbf{T}$ -algebra automorphism group of $\text{Rat}(\Gamma)$.*

Since φ is a $\frac{1}{\psi(1)}$ -expansive map $\Gamma' \rightarrow \Gamma$, for any $f \in \text{Rat}(\Gamma)$, slopes of f and $\psi(f)$ on intervals corresponding by φ coincide. Hence we obtain the following corollary:

Corollary 1.4. *Let Γ, Γ' be tropical curves. A semiring isomorphism $\psi : \text{Rat}(\Gamma) \rightarrow \text{Rat}(\Gamma')$ and the induced $\frac{1}{\psi(1)}$ -expansive map $\varphi : \Gamma' \rightarrow \Gamma$ map zeros (resp. poles) to zeros (resp. poles) and preserve their degrees.*

Note that we can restrict ourselves to $\mathbf{T}$ -algebra isomorphism setting. In this setting, $\psi(1)$ is always one, and thus we need not deal with r -expansive maps.

The rest of this paper is organized as follows. In Section 2, we give the definitions of semirings and algebras, tropical curves, rational functions and chip-firing moves on tropical curves, and finite harmonic morphisms between tropical curves. Section 3 contains proofs of all the assertions above. In that section, we also describe all semiring automorphisms of rational function semifields of all tropical curves; for all tropical curves Γ except for star-shaped tropical curves consisting of a finite number (at least one) of $[0, \infty]$, the semiring automorphism group of $\text{Rat}(\Gamma)$ coincides with the $\mathbf{T}$ -algebra automorphism group of $\text{Rat}(\Gamma)$ (Lemma 3.12 and Corollary 3.13).

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2. Preliminaries

In this section, we recall several basic definitions which we need later. We refer to [5] (resp. [19]) for an introduction to the theory of semirings (resp. tropical geometry) and employ definitions in [15] (resp. [11], [7] and [3]) related to semirings (resp. tropical curves, rational functions and finite harmonic morphisms) with slight modification partially.

2.1. Semirings and algebras

In this part, a *semiring* is a commutative semiring with the absorbing neutral element 0 for addition and the identity 1 for multiplication such that $0 \neq 1$. If every nonzero element of a semiring S is multiplicatively invertible, then S is called a *semifield*. A semiring S is *additively idempotent* if $x + x = x$ for any $x \in S$. An additively idempotent semiring S has a natural partial order, i.e., $x \geq y$ if and only if $x + y = x$ for $x, y \in S$.

A map $\varphi : S_1 \rightarrow S_2$ between semirings is a *semiring homomorphism* if for any $x, y \in S_1$,

$$\varphi(x + y) = \varphi(x) + \varphi(y), \quad \varphi(x \cdot y) = \varphi(x) \cdot \varphi(y), \quad \varphi(0) = 0, \quad \text{and} \quad \varphi(1) = 1.$$

A semiring homomorphism $\varphi : S_1 \rightarrow S_2$ is a *semiring isomorphism* if φ is bijective. A *semiring automorphism* of S is a semiring isomorphism $S \rightarrow S$.

Given a semiring homomorphism $\varphi : S_1 \rightarrow S_2$, we call the pair (S_2, φ) (for short, S_2) an S_1 -*algebra*. For a semiring S_1 , a map $\psi : (S_2, \varphi) \rightarrow (S'_2, \varphi')$ between S_1 -algebras is an S_1 -*algebra homomorphism* if ψ is a semiring homomorphism and $\varphi' = \psi \circ \varphi$. When there is no confusion, we write $\psi : S_2 \rightarrow S'_2$ simply. A bijective S_1 -algebra homomorphism $S_2 \rightarrow S'_2$ is a S_1 -*algebra isomorphism*. In addition, if $S_2 = S'_2$, then it is an S_1 -*algebra automorphism*.

The set $\mathbf{T} := \mathbf{R} \cup \{-\infty\}$ with two tropical operations:

$$a \oplus b := \max\{a, b\} \quad \text{and} \quad a \odot b := a + b,$$

where both a and b are in $\mathbf{T}$, becomes a semifield. Here, for any $a \in \mathbf{T}$, we handle $-\infty$ as follows:

$$a \oplus (-\infty) = (-\infty) \oplus a = a \quad \text{and} \quad a \odot (-\infty) = (-\infty) \odot a = -\infty.$$

This $\mathbf{T}$ is called the *tropical semifield*.

2.2. Tropical curves

In this part, a *graph* is an unweighted, undirected, finite, connected nonempty multigraph that may have loops. For a graph G , the set of vertices is denoted by $V(G)$ and the set of edges by $E(G)$. The *valence* of a vertex v of G is the number of edges incident to v , where each loop is counted twice. A vertex v of G is a *leaf end* if v has valence one. A *leaf edge* is an edge of G incident to a leaf end.

A *tropical curve* is the underlying topological space of the pair (G, l) of a graph G and a function $l : E(G) \rightarrow \mathbf{R}_{>0} \cup \{\infty\}$, where l can take the value ∞ only on leaf edges, together with an identification of each edge e of G with the closed interval $[0, l(e)]$. The interval $[0, \infty]$ is the one-point compactification of the interval $[0, \infty)$. We regard $[0, \infty]$ not just as a topological space but as an extended metric space. The distance between ∞ and any other point is infinite. When $l(e) = \infty$, the leaf end of e must be identified with ∞ . If $E(G) = \{e\}$ and $l(e) = \infty$, then we can identify either leaf ends of e with ∞ . When a tropical curve Γ is obtained from (G, l) , the pair (G, l) is called a *model* for Γ . There are many possible models for Γ . A model (G, l) is *loopless* if G is loopless. For a point x of a tropical curve Γ , if x is identified with ∞ , then x is called a *point at infinity*, else, x is called a *finite point*. Let Γ_∞ denote the set of all points at infinity of Γ . If x is a finite point, then the *valence* $\text{val}(x)$ is the number of connected components of $U \setminus \{x\}$ with any sufficiently small connected neighborhood U of x , if x is a point at infinity, then $\text{val}(x) := 1$. We remark that this “valence” is defined for a point of a tropical curve and the “valence” in the first paragraph of this subsection is defined for a vertex of a graph, and these are compatible with each other. We construct a model $(G_\circ, l_\circ)$ called the *canonical model* for Γ as follows. Generally, we define $V(G_\circ) := \{x \in \Gamma \mid \text{val}(x) \neq 2\}$ except for the following two cases. When Γ is homeomorphic to a circle S^1 , we define $V(G_\circ)$ as the set consisting of one arbitrary point of Γ . When Γ has the pair (T, l) as its model, where T is a tree consisting of three vertices and two edges and $l(E(T)) = \{\infty\}$, we define $V(G_\circ)$ as the set of two points at infinity and any finite point of Γ . The *genus* $g(\Gamma)$ of Γ is the first Betti number of Γ , which coincides with $\#E(G) - \#V(G) + 1$ for any model (G, l) for Γ . For a point x of Γ , a *half-edge* of x is a connected component of $U \setminus \{x\}$ with any connected neighborhood U of x which consists of only two valent points and x . We frequently identify a vertex (resp. an edge) of G with the corresponding point (resp. the corresponding closed subset) of Γ . The word “an edge of Γ ” means an edge of $G_\circ$. A tropical curve is *star-shaped* if it has a model whose underlying graph is a star graph with k leaves, i.e., the complete bipartite graph $K_{1,k}$ for some $k \in \mathbf{Z}_{>0}$.

2.3. Rational functions and chip-firing moves

Let Γ be a tropical curve. A continuous map $f : \Gamma \rightarrow \mathbf{R} \cup \{\pm\infty\}$ is a *rational function* on Γ if f is a constant function of $-\infty$ or a piecewise affine function with integer slopes, with a finite number of pieces and that can take the values $\pm\infty$ only at points at infinity. For a finite point x of Γ and a rational function $f \in \text{Rat}(\Gamma) \setminus \{-\infty\}$, for one outgoing direction at x , the *outgoing slope* of f at x is $\frac{f(y)-f(x)}{\text{dist}(x,y)}$ with a finite point y in a sufficiently small neighborhood of x in the direction. If x is a point at infinity, then we regard the outgoing slope of f at x as the slope of f from y to x times minus one, where y is a finite point on the leaf edge incident to x such that f has a constant slope on the interval (y, x) . In both cases, the definition of outgoing slope of f at x in the direction is independent of the choice of y . The point x is a *zero* (resp. *pole*) of f if the sign of the sum of outgoing slopes of f at x is positive (resp. negative). The absolute value of the sum is its *degree*. Let $\text{Rat}(\Gamma)$ denote the set of all rational functions on Γ . For rational functions $f, g \in \text{Rat}(\Gamma)$ and a point $x \in \Gamma \setminus \Gamma_\infty$, we define

$$(f \oplus g)(x) := \max\{f(x), g(x)\} \quad \text{and} \quad (f \odot g)(x) := f(x) + g(x).$$

We extend $f \oplus g$ and $f \odot g$ to points at infinity to be continuous on the whole of Γ . Then both are rational functions on Γ . Note that for any $f \in \text{Rat}(\Gamma)$, we have

$$f \oplus (-\infty) = (-\infty) \oplus f = f$$

and

$$f \odot (-\infty) = (-\infty) \odot f = -\infty.$$

Then $\text{Rat}(\Gamma)$ becomes a semifield with these two operations. Also, $\text{Rat}(\Gamma)$ becomes a $\mathbf{T}$ -algebra with the natural inclusion $\mathbf{T} \hookrightarrow \text{Rat}(\Gamma)$. Let $\text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma))$ (resp. $\text{Aut}_{\mathbf{T}}(\text{Rat}(\Gamma))$) denote the set of all semiring (resp. $\mathbf{T}$ -algebra) automorphisms of $\text{Rat}(\Gamma)$. Then both $\text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma))$ and $\text{Aut}_{\mathbf{T}}(\text{Rat}(\Gamma))$ have a group structure. Note that for $f, g \in \text{Rat}(\Gamma)$, the equality $f = g$ means that $f(x) = g(x)$ for any $x \in \Gamma$.

Let Γ_1 be a closed subset of a tropical curve Γ with a finite number of connected components which has no connected components consisting of only a point at infinity, and l a positive number or infinity. The *chip-firing move* by Γ_1 and l is defined as the function $\text{CF}(\Gamma_1, l)(x) := -\min\{\text{dist}(\Gamma_1, x), l\}$ with $x \in \Gamma$. By definition, $\text{CF}(\Gamma_1, l)$ has only zero, one and minus one as its slopes and a finite number of pieces and is continuous, hence it is a rational function on Γ . The notion of chip-firing moves is introduced from the graph theory as an analogue of “chip-firing games” (for example, see the survey article [9]) and the definition above seems to be given by Haase–Musiker–Yü in [7].

Example 2.1. Let $\Gamma := [0, 4]$, $x := 1$, $y := 2$ and $z := 3$. Then $\text{CF}(\{y\}, 2)$ has slope one from 0 to y and slope minus one from y to 4 and $\text{CF}(\{y\}, 2)(y) = 0$. Also $\text{CF}(\{y\}, 1)$ has slope zero on $[0, x]$ and $[z, 4]$ and slope one from x to y and slope minus one from y to z and $\text{CF}(\{y\}, 1)(y) = 0$. The chip-firing move $\text{CF}(\{x, z\}, 1)$ has slope one from 0 to x and from y to z and slope minus one from x to y and from z to 4 and $\text{CF}(\{x, z\}, 1)(x) = \text{CF}(\{x, z\}, 1)(z) = 0$. On $[x, z]$, the chip-firing move $\text{CF}([x, z], 1)$ is constant zero and has slope one from 0 to x and slope minus one from z to 4.

2.4. Finite harmonic morphisms

Let $\varphi : \Gamma \rightarrow \Gamma'$ be a continuous map between tropical curves. The map φ is a *finite harmonic morphism* if there exist loopless models (G, l) and (G', l') for Γ and Γ' , respectively, such that (1) $\varphi(V(G)) \subset V(G')$ holds, (2) $\varphi(E(G)) \subset E(G')$ holds, (3) for any edge e of G , there exists a positive integer $\deg_e(\varphi)$ such that for

any points x, y of e , the equality $\text{dist}_{\varphi(e)}(\varphi(x), \varphi(y)) = \deg_e(\varphi) \cdot \text{dist}_e(x, y)$ holds, where $\text{dist}_{\varphi(e)}(\varphi(x), \varphi(y))$ (resp. $\text{dist}_e(x, y)$) denotes the distance between $\varphi(x)$ and $\varphi(y)$ in $\varphi(e)$ (resp. x and y in e), and (4) for every vertex v of G , the sum $\sum_{e \in E(G): e \mapsto e', v \in e} \deg_e(\varphi)$ is independent of the choice of $e' \in E(G')$ incident to $\varphi(v)$. This sum is denoted by $\deg_v(\varphi)$. Then the sum $\sum_{v \in V(G): v \mapsto v'} \deg_v(\varphi)$ is independent of the choice of a vertex v' of G' , and is called the *degree* of φ , written by $\deg(\varphi)$. If both Γ and Γ' are singletons, i.e., consist of only one point, we regard φ as a finite harmonic morphism that can have any positive integer as its degree. A finite harmonic morphism of degree one is bijective and its inverse is also a finite harmonic morphism of degree one, hence it is an *isomorphism* of tropical curves. For a tropical curve Γ , an isomorphism $\Gamma \rightarrow \Gamma$ is an *automorphism* of Γ . Note that if $\varphi \circ \psi$ is a composition of finite harmonic morphisms, then it is also a finite harmonic morphism of degree $\deg(\varphi) \cdot \deg(\psi)$, and thus tropical curves and finite harmonic morphisms between them make a category. Let $\text{Aut}(\Gamma)$ denote the set of all automorphisms of Γ and then $\text{Aut}(\Gamma)$ becomes a group by compositions of maps.

Remark 2.2. This $\text{Aut}(\Gamma)$ coincides with the set of all continuous maps $\Gamma \rightarrow \Gamma$ whose restrictions on $\Gamma \setminus \Gamma_\infty$ are isometries $\Gamma \setminus \Gamma_\infty \rightarrow \Gamma \setminus \Gamma_\infty$.

3. Main results

In this section, we will prove all assertions in Section 1.

We start our consideration from semiring automorphisms of $\mathbf{T}$. For an $r \in \mathbf{R}_{>0}$, we call the map $f_r : \mathbf{T} \rightarrow \mathbf{T}; t \mapsto r \cdot t$ the *expansive map*, where we define $r \cdot (-\infty) = -\infty$.

Proposition 3.1. For $r \in \mathbf{R}_{>0}$, the r -expansive map $f_r : \mathbf{T} \rightarrow \mathbf{T}; t \mapsto r \cdot t$ is a semiring automorphism.

Proof. Clearly, we have

$$f_r(0) = r \cdot 0 = 0$$

and

$$f_r(-\infty) = r \cdot (-\infty) = -\infty.$$

We also have

$$f_r(a \odot b) = r \cdot (a \odot b) = r \cdot (a + b) = r \cdot a + r \cdot b = f_r(a) \odot f_r(b).$$

When $a \oplus b = b$, since $a \leq b$ and $r > 0$, we have $r \cdot a \leq r \cdot b$. Hence,

$$f_r(a \oplus b) = f_r(b) = r \cdot b = (r \cdot a) \oplus (r \cdot b) = f_r(a) \oplus f_r(b)$$

hold. By a similar argument in the case of $a \oplus b = a$, the map f_r is a semiring homomorphism.

Assume that $f_r(a) = f_r(b)$. If $f_r(a) = f_r(b) = -\infty$, then $r \cdot a = r \cdot b = -\infty$. Since $r \neq 0, -\infty$, we have $a = b = -\infty$. If $f_r(a) = f_r(b) \neq -\infty$, then $r \cdot a = r \cdot b \in \mathbf{R}$. Since $r \neq 0$, we have $a = b$. Therefore f_r is injective.

For any $a \in \mathbf{R}$, since $r \neq 0$, we have $a = r \cdot \frac{a}{r} = f_r(\frac{a}{r})$. This means that f_r is surjective.

In conclusion, f_r is a semiring automorphism. $\square$

Does the converse of Proposition 3.1 hold? The answer is yes by [16, Corollary 3.16]. Here we give another proof for the readability.

Lemma 3.2. Let S_1, S_2 be additively idempotent semirings. A semiring homomorphism $f : S_1 \rightarrow S_2$ is order preserving.

Proof. For $a, b \in S_1$, $a \leq b$ if and only if $a + b = b$. Since f is a semiring homomorphism, if $a \leq b$, then we have $f(a) + f(b) = f(a + b) = f(b)$. This means that $f(a) \leq f(b)$. $\square$

Proposition 3.3. For a semiring automorphism $\psi : \mathbf{T} \rightarrow \mathbf{T}$, $\psi(1) \in \mathbf{R}_{>0}$ and ψ is the $\psi(1)$ -expansive map $\mathbf{T} \rightarrow \mathbf{T}; t \mapsto \psi(1) \cdot t$.

Proof. Since ψ is a semiring homomorphism, for any $n \in \mathbf{Z} \setminus \{0\}$ and $t \in \mathbf{R}$, we have $\psi(n \cdot t) = \psi(t^{\odot n}) = \psi(t)^{\odot n} = n \cdot \psi(t)$.

Assume that for any $r \in \mathbf{R}_{>0}$, this ψ is not the r -expansive map. Then there exist $t < t' \in \mathbf{R} \setminus \{0\}$ such that $\frac{t'}{t} \neq \frac{\psi(t')}{\psi(t)}$. Note that since $t \neq 0$, so is $\psi(t)$.

Suppose that $t < t' < 0$. By Lemma 3.2, we have $\psi(t) < \psi(t') < 0$. Thus the signs of ratios $\frac{t'}{t}$ and $\frac{\psi(t')}{\psi(t)}$ coincide. Since $\frac{t'}{t} = \frac{-t'}{-t}$ and $\frac{\psi(t')}{\psi(t)} = \frac{-\psi(t')}{-\psi(t)} = \frac{\psi(t')^{\odot(-1)}}{\psi(t)^{\odot(-1)}} = \frac{\psi(-t')}{\psi(-t)}$, we have $\frac{-t'}{-t} \neq \frac{\psi(-t')}{\psi(-t)}$. Hence by replacing t, t' with $-t, -t'$, respectively, we can assume that $0 < t < t'$ in this case.

Suppose that $t < 0 < t'$. If $\frac{t'}{t} = -1$, then $t' = -t$. Therefore, $\frac{\psi(t')}{\psi(t)} = \frac{\psi(-t)}{\psi(t)} = \frac{-\psi(t)}{\psi(t)} = -1$, and this is a contradiction. Thus $t' \neq -t$, and $0 < -t < t'$ or $0 < t' < -t$. Hence $\frac{t'}{-t} = -\frac{t'}{t} \neq -\frac{\psi(t')}{\psi(t)} = \frac{\psi(t')}{-\psi(t)} = \frac{\psi(t')}{\psi(-t)}$. From these arguments, it is enough to consider when $0 < t < t'$.

Assume that $0 < t < t'$. Let $a := \frac{t'}{t}$. Then $a > 1$. Since $0 < \psi(t) < \psi(t')$ by Lemma 3.2, if b denotes the ratio $\frac{\psi(t')}{\psi(t)}$, then $b > 1$. By assumption, $a \neq b$.

Assume that $a > b$. By the density of rational numbers, there exists a rational number $c \in \mathbf{Q}_{>1}$ such that $1 < b < c < a$. By the Archimedean property of the real numbers, there exists a positive integer $m \in \mathbf{Z}_{\geq 1}$ such that $m \cdot c > a$. We can assume that $m \cdot c \in \mathbf{Z}_{>1}$. Since $t > 0$, we have

$$0 < t < b \cdot t < c \cdot t < a \cdot t = t' < m \cdot c \cdot t.$$

By Lemma 3.2, we have

$$\begin{aligned} 0 < \psi(t) < \psi(b \cdot t) < \psi(c \cdot t) < \psi(a \cdot t) &= \psi(t') = b \cdot \psi(t) \\ < \psi(m \cdot c \cdot t) &= m \cdot \psi(c \cdot t) = m \cdot c \cdot \psi(t). \end{aligned}$$

From these arguments, we have $\psi(c \cdot t) = c \cdot \psi(t)$ and $c \cdot \psi(t) < b \cdot \psi(t)$. Since $0 < \psi(t)$, this means that $c < b$, a contradiction. Hence $a \geq b$. By a similar argument, we have $a \leq b$. Therefore $a = b$ holds but this contradicts $a \neq b$. Thus there are no such $t < t'$. This means that for any $t < t' \in \mathbf{R} \setminus \{0\}$, $\frac{t'}{t} = \frac{\psi(t')}{\psi(t)}$, and thus $\frac{\psi(t)}{t} = \frac{\psi(t')}{t'}$. Hence $\frac{\psi(t)}{t} = \frac{\psi(1)}{1} > 0$. In conclusion, ψ is the $\psi(1)$ -expansive map. $\square$

Corollary 3.4. The semiring automorphism group of $\mathbf{T}$ is isomorphic to the group consisting of the r -expansive maps of $\mathbf{T}$ for all $r \in \mathbf{R}_{>0}$.

In what follows, we will consider semiring isomorphisms between rational function semifields of tropical curves.

Lemma 3.5. Let Γ, Γ' be tropical curves. If $\psi : \text{Rat}(\Gamma) \rightarrow \text{Rat}(\Gamma')$ is a semiring isomorphism, then $\psi(\mathbf{T}) = \mathbf{T}$.

Proof. Since ψ is a semiring homomorphism, we have $\psi(-\infty) = -\infty$. Let $t \in \mathbf{R}$ and $n \in \mathbf{Z}_{>0}$. Then $(\frac{t}{n})^{\odot n} = t$. Thus we have

$$\psi(t) = \psi\left(\left(\frac{t}{n}\right)^{\odot n}\right) = \psi\left(\frac{t}{n}\right)^{\odot n} = n \cdot \psi\left(\frac{t}{n}\right).$$

Therefore each slope of $\psi\left(\frac{t}{n}\right)$ is $\frac{1}{n}$ times that of $\psi(t)$. Since each rational function on a tropical curve other than $-\infty$ has only integer slopes, $\psi(t)$ must be in $\mathbf{T}$. Thus $\psi(\mathbf{T}) \subset \mathbf{T}$. Similarly, we have $\psi^{-1}(\mathbf{T}) \subset \mathbf{T}$. Hence $\psi(\mathbf{T}) = \mathbf{T}$. $\square$

Remark 3.6. By Proposition 3.3 and Lemma 3.5, the restriction $\psi|_{\mathbf{T}}$ is the $\psi(1)$ -expansive map $\mathbf{T} \rightarrow \mathbf{T}$, where ψ is as in Lemma 3.5. In particular, if ψ is a $\mathbf{T}$ -algebra isomorphism, then $\psi(1) = 1$.

Let Γ, Γ' be tropical curves and $\psi : \text{Rat}(\Gamma) \rightarrow \text{Rat}(\Gamma')$ a semiring isomorphism. Let $r := \psi(1)$. The following lemma is easy but fundamental.

Lemma 3.7. *For any $f \in \text{Rat}(\Gamma)$, the following hold:*

- (1) $r \cdot \max\{f(x) \mid x \in \Gamma\} = \max\{\psi(f)(x') \mid x' \in \Gamma'\}$, and
- (2) $r \cdot \min\{f(x) \mid x \in \Gamma\} = \min\{\psi(f)(x') \mid x' \in \Gamma'\}$.

Proof. If $f \in \mathbf{T}$, the assertions are clear.

Assume that f is not a constant function. Let a be the maximum value of f . In this case, a is in $\mathbf{R} \cup \{\infty\}$.

Assume $a \in \mathbf{R}$. For $b \in \mathbf{T}$, we have

$$f \oplus b \begin{cases} = b & \text{if } b \geq a, \\ \neq b & \text{if } b < a. \end{cases}$$

Therefore we have

$$\psi(f) \oplus r \cdot b = \psi(f) \oplus \psi(b) = \psi(f \oplus b) \begin{cases} = \psi(b) = r \cdot b & \text{if } b \geq a, \\ \neq \psi(b) = r \cdot b & \text{if } b < a. \end{cases}$$

Thus the maximum value of $\psi(f)$ is $r \cdot a$.

Assume $a = \infty$. Then for any $t \in \mathbf{T}$, we have $f \oplus t \neq t$. Thus

$$\psi(f) \oplus r \cdot t = \psi(f) \oplus \psi(t) = \psi(f \oplus t) \neq \psi(t) = r \cdot t$$

hold. This means that the maximum value of $\psi(f)$ is ∞ since tropical curves are compact.

For the minimum values of f and $\psi(f)$, we can obtain the conclusion by applying the maximum value case for $f^{\odot(-1)} = -f$ and $\psi(f^{\odot(-1)}) = -\psi(f)$ since

$$\min\{f(x) \mid x \in \Gamma\} = -\max\{-f(x) \mid x \in \Gamma\}$$

and

$$\min\{\psi(f)(x') \mid x' \in \Gamma'\} = -\max\{-\psi(f)(x') \mid x' \in \Gamma'\}. \quad \square$$

Lemma 3.8 is our key lemma to prove Theorem 1.1.

Lemma 3.8. (1) *For any $x \in \Gamma \setminus \Gamma_{\infty}$, there exist $\varepsilon > 0$ and a unique point $x' \in \Gamma' \setminus \Gamma'_{\infty}$ such that $\psi(\text{CF}(\{x\}, \varepsilon)) = \text{CF}(\{x'\}, r \cdot \varepsilon)$. Furthermore, $\text{val}(x) = \text{val}(x')$ holds (Fig. 1).*

(2) *For any $x \in \Gamma_{\infty}$, there exists a finite point y on the leaf edge e incident to x and there exists a unique point $x' \in \Gamma'_{\infty}$, such that the point $y' \in \Gamma' \setminus \Gamma'_{\infty}$ corresponding to y (in the meaning of (1)) is on the leaf edge incident to x' and $\psi(\text{CF}(\Gamma \setminus (y, x], \infty)) = \text{CF}(\Gamma' \setminus (y', x'], \infty)$ (Fig. 2).*

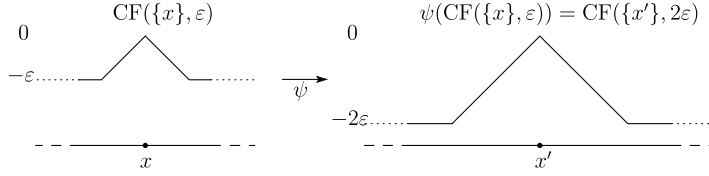


Fig. 1. An example of Lemma 3.8(1). The figure stands for neighborhoods of x and x' in the case of $\text{val}(x) = 2$ and $r = \psi(1) = 2$.

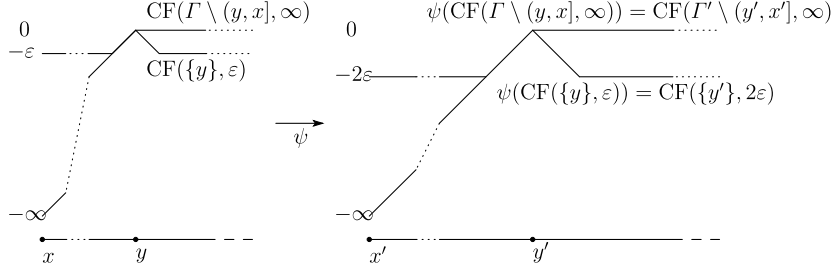


Fig. 2. An example of Lemma 3.8(2). The figure stands for Γ, Γ' around $(y, x], (y', x']$ in the case of $r = \psi(1) = 2$ for a sufficiently small positive number ε .

In the proof of Lemma 3.8, we will consider several times rational functions of the form $(f^{\odot(-1)} \oplus a)^{\odot(-1)}$ for a rational function f other than $-\infty$ on a tropical curve Γ and $a \in \mathbf{R}$. It is equal to the minimum of f and $-a$, i.e., for any $x \in \Gamma$,

$$(f^{\odot(-1)} \oplus a)^{\odot(-1)}(x) = \min\{f(x), -a\},$$

where we define $\min\{\infty, t\} = t$ and $\min\{-\infty, t\} = -\infty$ for any $t \in \mathbf{R}$. Also, if

$$f = \bigoplus_{i=1}^n g_i$$

and for any j ,

$$f \neq \bigoplus_{i \neq j, i=1}^n g_i$$

hold, then we call this finite sum $\bigoplus_{i=1}^n g_i$ an *irredundant representation* of f . The image of an irredundant representation by a semiring isomorphism is also an irredundant representation. In particular, if f is a chip-firing move of the form of Lemma 3.8(1) on Γ and if $\bigoplus_{i=1}^n g_i$ is an irredundant representation of f satisfying the following condition $(\star)$, then $n \leq \text{val}(x)$.

- $(\star) \left\{ \begin{array}{l} (1) \text{ The maximum value of each } g_i \text{ is zero.} \\ (2) \text{ The minimum value of each } g_i \text{ is } -\varepsilon. \\ (3) \text{ Each } g_i \text{ has a unique zero on the closure of each half-edge of } x \text{ in the } \varepsilon\text{-neighborhood of } x. \end{array} \right.$

Similarly, if f is a chip-firing move of the form of Lemma 3.8(2) on Γ and if $\bigoplus_{i=1}^n g_i$ is an irredundant representation of $f^{\odot(-1)}$ satisfying the following condition $(\star\star)$, then $n = \text{val}(x) = 1$.

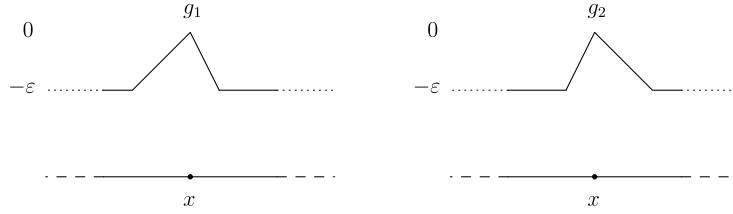


Fig. 3. An example of rational functions g_1, g_2 whose (tropical) sum is an irredundant representation of the chip-firing move of the left figure of Fig. 1 satisfying the condition $(\star)$.

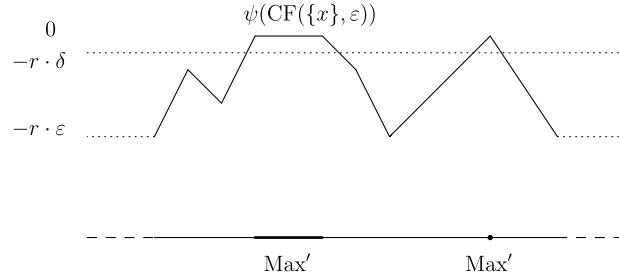


Fig. 4. An example of the first paragraph of the proof of Lemma 3.8.

$$(\star\star) \begin{cases} (1) \text{ The maximum value of each } g_i \text{ is } \infty. \\ (2) \text{ The minimum value of each } g_i \text{ is zero.} \\ (3) \text{ Each } g_i \text{ has a unique zero on } [y, x]. \end{cases}$$

Proof of Lemma 3.8. We shall prove (1). For any $\varepsilon > 0$, we consider $\psi(\text{CF}(\{x\}, \varepsilon))$. By Lemma 3.7, the maximum value of $\psi(\text{CF}(\{x\}, \varepsilon))$ is zero. Let Max' denote the set of all points of Γ' where $\psi(\text{CF}(\{x\}, \varepsilon))$ attains the maximum value 0. Since $\text{CF}(\{x\}, \varepsilon)$ is not a constant function, $\psi(\text{CF}(\{x\}, \varepsilon))$ is also not a constant function. Therefore the boundary set $\partial \text{Max}'$ of Max' is a non-empty finite set. For δ such that $0 < \delta < \varepsilon$, since

$$\text{CF}(\{x\}, \delta) = \text{CF}(\{x\}, \varepsilon) \oplus (-\delta),$$

we have

$$\psi(\text{CF}(\{x\}, \delta)) = \psi(\text{CF}(\{x\}, \varepsilon)) \oplus (-r \cdot \delta).$$

Therefore, if necessary, by replacing ε with δ , we may assume that $\psi(\text{CF}(\{x\}, \varepsilon))$ has a unique zero on the closure of each half-edge of each point $x' \in \partial \text{Max}'$ in the $(r \cdot \varepsilon)$ -neighborhood of Max' . Also, when $\psi(\text{CF}(\{x\}, \varepsilon))$ has a pole outside of Max' , by replacing ε with the $\frac{1}{r}$ times the maximum value of $\psi(\text{CF}(\{x\}, \varepsilon))$ on such poles as above δ , we may assume that there are no such poles. See Fig. 4. Hereafter, we assume these two conditions.

Assume that $\# \text{Max}'$ is infinite. Since $\psi(\text{CF}(\{x\}, \varepsilon))$ is a rational function, Max' has a finite number of connected components. Thus Max' has a connected component that is an infinite set. For such a connected component Γ'_1 , let $y' \in \Gamma'_1 \setminus (\partial \Gamma'_1 \cup \Gamma'_\infty)$. Let $0 < \delta' \leq \min\{\text{dist}(y', z') \mid z' \in \partial \Gamma'_1\}$. Let f denote the rational function

$$\psi^{-1}(\psi(\text{CF}(\{x\}, \varepsilon)) \odot \text{CF}(\{y'\}, \delta') \odot \delta').$$

Since

$$\left(\psi(\text{CF}(\{x\}, \varepsilon)) \odot \text{CF}(\{y'\}, \delta') \odot \delta'^{\odot(-1)} \oplus 0 \right)^{\odot(-1)} = \psi(\text{CF}(\{x\}, \varepsilon)),$$

we have

$$\begin{aligned} & \text{CF}(\{x\}, \varepsilon) \\ &= \psi^{-1} \left(\left[\psi(\text{CF}(\{x\}, \varepsilon)) \odot \text{CF}(\{y'\}, \delta') \odot \delta'^{\odot(-1)} \oplus 0 \right]^{\odot(-1)} \right) \\ &= \left(\psi^{-1} \left(\psi(\text{CF}(\{x\}, \varepsilon)) \odot \text{CF}(\{y'\}, \delta') \odot \delta'^{\odot(-1)} \oplus 0 \right) \right)^{\odot(-1)} \\ &= \left(f^{\odot(-1)} \oplus 0 \right)^{\odot(-1)}. \end{aligned}$$

As the maximum value of f is $\frac{\delta'}{r}$ by Lemma 3.7, by the shape of $\text{CF}(\{x\}, \varepsilon)$, the equality $\text{CF}(\{x\}, \varepsilon) = (f^{\odot(-1)} \oplus 0)^{\odot(-1)}$ means that f takes the maximum value $\frac{\delta'}{r}$ at and only at x and at most zero at any other point, which cannot occur since a rational function is continuous. Hence Max' is a finite set.

If $\text{Max}' \cap I'_\infty \neq \emptyset$, then for any $x' \in \text{Max}' \cap I'_\infty$, the rational function $\psi(\text{CF}(\{x\}, \varepsilon))$ takes the maximum value 0 at x' and has x' as a pole since Max' is finite. However, by the definition of rational functions, this does not occur. Thus $\text{Max}' \cap I'_\infty$ is empty.

For any $x' \in \text{Max}'$, let $g'_{x'}$ be the rational function on I' that coincides with $\psi(\text{CF}(\{x\}, \varepsilon))$ in the $(r \cdot \varepsilon)$ -neighborhood of x' and is $-r \cdot \varepsilon$ on other points. Then we have

$$\psi(\text{CF}(\{x\}, \varepsilon)) = \bigoplus_{x' \in \text{Max}'} g'_{x'}.$$

The right-hand side is an irredundant representation. Hence we have

$$\text{CF}(\{x\}, \varepsilon) = \bigoplus_{x' \in \text{Max}'} \psi^{-1}(g'_{x'}).$$

This right-hand side is also an irredundant representation. If necessary, by replacing ε with a smaller positive number, we may assume that this irredundant representation satisfies the condition $(\star)$. Thus the number of elements of Max' is at most the valence of x . In particular, $\text{val}(x) = 1$ implies $\# \text{Max}' = 1$.

We will show that if $\text{val}(x) = 1$, then for the unique element x' of Max' , the equalities $\text{val}(x') = 1$ and $\psi(\text{CF}(\{x\}, \varepsilon)) = \text{CF}(\{x'\}, r \cdot \varepsilon)$ hold. Assume that $\text{val}(x') \geq 2$. Let $m = \text{val}(x')$ and $h'_1, \dots, h'_m$ be the half-edges of x' in the $(r \cdot \varepsilon)$ -neighborhood of x' . For each $i = 1, \dots, m$, let g'_i be the rational function on I' defined as follows: g'_i coincides with $\psi(\text{CF}(\{x\}, \varepsilon))$ on h'_i , and, g'_i has the slope $s(< 0)$ of $\psi(\text{CF}(\{x\}, \varepsilon))$ minus one on the intersection of h'_j and the $(r \cdot \varepsilon/(|s| + 1))$ -neighborhood of x' for any $j \neq i$, and, g'_i takes the minimum value $-r \cdot \varepsilon$ on other points (cf. Fig. 3 for the construction of g'_i s). Then we have

$$\psi(\text{CF}(\{x\}, \varepsilon)) = \bigoplus_{i=1}^m g'_i.$$

The right-hand side is an irredundant representation. Hence we obtain the irredundant representation $\bigoplus_{i=1}^m \psi^{-1}(g'_i)$ of $\text{CF}(\{x\}, \varepsilon)$ consisting of m elements. Again, if necessary, by replacing ε with a smaller positive number, we may assume that this irredundant representation satisfies the condition $(\star)$. This means that $m \leq 1$, a contradiction. Thus $\text{val}(x')$ must be one. Also if $\psi(\text{CF}(\{x\}, \varepsilon))$ is not $\text{CF}(\{x'\}, r \cdot \varepsilon)$, then

$$\psi(\text{CF}(\{x\}, \varepsilon)) \oplus \text{CF}(\{x'\}, r \cdot \varepsilon) = \text{CF}(\{x'\}, r \cdot \varepsilon).$$

Then we have

$$\text{CF}(\{x\}, \varepsilon) \neq \psi^{-1}(\text{CF}(\{x'\}, r \cdot \varepsilon))$$

and

$$\text{CF}(\{x\}, \varepsilon) \oplus \psi^{-1}(\text{CF}(\{x'\}, r \cdot \varepsilon)) = \psi^{-1}(\text{CF}(\{x'\}, r \cdot \varepsilon)).$$

However, since $\psi^{-1}(\text{CF}(\{x'\}, r \cdot \varepsilon))$ attains the maximum value 0 at x and only at x , and since the minimum value of $\psi^{-1}(\text{CF}(\{x'\}, r \cdot \varepsilon))$ is $-\varepsilon$ by Lemma 3.7, it does not occur. Therefore $\psi(\text{CF}(\{x\}, \varepsilon))$ must be $\text{CF}(\{x'\}, r \cdot \varepsilon)$.

Assume that when $\text{val}(x) = k$, the assertion holds. Consider the case $\text{val}(x) = k + 1$.

Suppose that Max' contains a point y' of valence at most k . If necessary, by replacing ε with a sufficiently small positive number, by assumption, there exists a unique point $y \in \Gamma \setminus \Gamma_\infty$ such that

$$\psi^{-1}(\text{CF}(\{y'\}, r \cdot \varepsilon)) = \text{CF}(\{y\}, \varepsilon)$$

and $k \geq \text{val}(y') = \text{val}(y)$. Thus $x \neq y$. On the other hand, since

$$g'_{y'} \oplus \text{CF}(\{y'\}, r \cdot \varepsilon) = \text{CF}(\{y'\}, r \cdot \varepsilon)$$

and

$$\bigoplus_{x' \in \text{Max}'} g'_{x'} = \psi(\text{CF}(\{x\}, \varepsilon)),$$

we have

$$\psi^{-1}(g'_{y'}) \oplus \text{CF}(\{y\}, \varepsilon) = \text{CF}(\{y\}, \varepsilon)$$

and

$$\bigoplus_{x' \in \text{Max}'} \psi^{-1}(g'_{x'}) = \text{CF}(\{x\}, \varepsilon).$$

By the latter equality, $\psi^{-1}(g'_{y'})$ must attain the maximum value 0 at x and only at x . However, by the former equality, this means $x = y$, a contradiction. Therefore Max' has no points of valence at most k .

Assume that $\#\text{Max}' \geq 2$. For any $x' \in \text{Max}'$, let $g'_{x',1}, \dots, g'_{x',\text{val}(x')}$ be the rational functions on Γ' defined as the above g'_i 's which is not constant only on the $(r \cdot \varepsilon)$ -neighborhood of x' . Then we have

$$\psi(\text{CF}(\{x\}, \varepsilon)) = \bigoplus_{x' \in \text{Max}'} \bigoplus_{i=1}^{\text{val}(x')} g'_{x',i}.$$

The right-hand side is an irredundant representation. This means that if ε is sufficiently small, then $\text{CF}(\{x\}, \varepsilon)$ has an irredundant representation

$$\bigoplus_{x' \in \text{Max}'} \bigoplus_{i=1}^{\text{val}(x')} \psi^{-1}(g'_{x',i})$$

consisting of $\sum_{x' \in \text{Max}'} \text{val}(x') > \text{val}(x)$ elements and satisfying the condition $(\star)$. This is a contradiction. Thus $\# \text{Max}'$ must be one. Let x' be the unique point in Max' . If $\text{val}(x) < \text{val}(x')$, then by a similar argument above, we have a contradiction. Therefore $\text{val}(x) = \text{val}(x')$. Also by a similar argument in the case of $\text{val}(x) = 1$, the rational function $\psi(\text{CF}\{x\}, \varepsilon)$ is equal to $\text{CF}(\{x'\}, r \cdot \varepsilon)$. Hence we have the conclusion.

We shall prove (2). Let y be a point of $e \setminus I'_\infty$. By Lemma 3.7, the maximum (resp. minimum) value of $\psi(\text{CF}(I' \setminus (y, x], \infty))$ is 0 (resp. $-\infty$). Thus if Min' denotes the set of all points of I' where $\psi(\text{CF}(I' \setminus (y, x], \infty))$ attains the minimum value $-\infty$, then $\text{Min}' \subset I'_\infty$. Hence Min' is finite. For a sufficiently large number $a > 0$, there exists a unique point $z \in (y, x)$ such that

$$\left\{ (-a) \odot \text{CF}(I' \setminus (y, x], \infty)^{\odot(-1)} \oplus 0 \right\}^{\odot(-1)} = \text{CF}(I' \setminus (z, x], \infty).$$

Thus we have

$$\left\{ (-r \cdot a) \odot \psi(\text{CF}(I' \setminus (y, x], \infty))^{\odot(-1)} \oplus 0 \right\}^{\odot(-1)} = \psi(\text{CF}(I' \setminus (z, x], \infty)).$$

Hence, if necessary, by replacing y with z , we may assume the following: $\psi(\text{CF}(I' \setminus (y, x], \infty))$ has no zeros on $I' \setminus I'_\infty$, has a unique pole on the leaf edge incident to each element of Min' , and has no other poles. Hereafter, we assume these three conditions. If P' denotes the set of all poles of $\psi(\text{CF}(I' \setminus (y, x], \infty))$, then there exists a natural one-to-one correspondence between P' and Min' . Let $y' \in P'$ correspond to $x' \in \text{Min}'$. Then there exists a unique positive integer $n_{y'}$ such that

$$\begin{aligned} & \text{CF}(I' \setminus (y', x'], \infty)^{\odot n_{y'}} \\ &= \psi(\text{CF}(I' \setminus (y, x], \infty)) \oplus \text{CF}(I' \setminus (y', x'], \infty)^{\odot n_{y'}} \end{aligned}$$

and

$$\begin{aligned} & \text{CF}(I' \setminus (y', x'], \infty)^{\odot(n_{y'}+1)} \\ & \neq \psi(\text{CF}(I' \setminus (y, x], \infty)) \oplus \text{CF}(I' \setminus (y', x'], \infty)^{\odot(n_{y'}+1)}. \end{aligned}$$

By considering the images of these equalities by ψ^{-1} , we have $n_{y'} = 1$. Let $g'_{y'}$ be $\text{CF}(I' \setminus (y', x'], \infty)^{\odot(-1)}$. Then $\bigoplus_{y' \in P'} g'_{y'}$ is an irredundant representation of $\psi(\text{CF}(I' \setminus (y, x], \infty))^{\odot(-1)}$. If necessary, by replacing y with $z \in (y, x)$ such that $\text{dist}(y, z)$ is sufficiently large, we may assume that the irredundant representation $\bigoplus_{y' \in P'} \psi^{-1}(g'_{y'})$ of $\text{CF}(I' \setminus (y, x], \infty)^{\odot(-1)}$ satisfies the condition $(\star\star)$. Hence we have $\#P' = \#\text{Min}' = 1$. Therefore, we have

$$\psi(\text{CF}(I' \setminus (y, x], \infty)) = \text{CF}(I' \setminus (y', x'], \infty).$$

We will show that y corresponds to the unique pole $y' \in P'$. By (1), for a sufficiently small positive number ε , there exists a unique $z' \in I' \setminus I'_\infty$ such that

$$\psi(\text{CF}(\{y\}, \varepsilon)) = \text{CF}(\{z'\}, r \cdot \varepsilon).$$

Since

$$\begin{aligned} & \text{CF}(I' \setminus (y', x'], \infty) \oplus (-r \cdot \varepsilon) \\ &= \psi(\text{CF}(I' \setminus (y, x], \infty)) \oplus (-r \cdot \varepsilon) \\ &= \psi(\text{CF}(I' \setminus (y, x], \infty) \oplus (-\varepsilon)) \end{aligned}$$

$$\begin{aligned}
&= \psi(\text{CF}(\Gamma \setminus (y, x], \infty) \oplus \text{CF}(\{y\}, \varepsilon)) \\
&= \text{CF}(\Gamma' \setminus (y', x'], \infty) \oplus \text{CF}(\{z'\}, r \cdot \varepsilon),
\end{aligned}$$

we have $z' \notin (y', x']$. Suppose that $y' \neq z'$. If necessary, by replacing ε with a positive number at most $\frac{1}{r} \cdot \text{dist}(y', z')$, we obtain

$$\begin{aligned}
&(-r \cdot \varepsilon) \odot \text{CF}(\Gamma' \setminus (y', x'], \infty) \\
&= \left(\{ \text{CF}(\Gamma' \setminus (y', x'], \infty) \odot \text{CF}(\{z'\}, r \cdot \varepsilon) \}^{\odot(-1)} \oplus r \cdot \varepsilon \right)^{\odot(-1)}.
\end{aligned}$$

On the other hand, the image of this equality by ψ^{-1} is

$$\begin{aligned}
&(-\varepsilon) \odot \text{CF}(\Gamma \setminus (y, x], \infty) \\
&= \left(\{ \text{CF}(\Gamma \setminus (y, x], \infty) \odot \text{CF}(\{y\}, \varepsilon) \}^{\odot(-1)} \oplus \varepsilon \right)^{\odot(-1)},
\end{aligned}$$

and this is a contradiction. In conclusion, y' must be z' . $\square$

Now, we can prove Theorem 1.1.

Proof of Theorem 1.1. Lemma 3.8 induces a map $\varphi' : \Gamma \rightarrow \Gamma'; x \mapsto x'$. We will show that this φ' is a $\psi(1)$ -expansive map. Since ψ is bijective, φ' is also bijective. Put $r := \psi(1)$. For any $x \in \Gamma \setminus \Gamma_\infty$ and a positive number ε obtained by Lemma 3.8(1), let U be the ε -neighborhood of x . For any $y \in U \setminus \{x\}$, let $d = \text{dist}(x, y)$. Let δ be a positive number for y obtained by Lemma 3.8(1). If necessary, we replace ε with a positive number between d and $d + \delta$. Then for $a \in \mathbf{T}$, we have

$$\text{CF}(\{x\}, \varepsilon) \oplus \text{CF}(\{y\}, \delta) \odot a \begin{cases} = \text{CF}(\{x\}, \varepsilon) & \text{if } a \leq -d, \\ \neq \text{CF}(\{x\}, \varepsilon) & \text{if } a > -d. \end{cases}$$

The images of these by ψ are

$$\text{CF}(\{\varphi'(x)\}, r \cdot \varepsilon) \oplus \text{CF}(\{\varphi'(y)\}, r \cdot \delta) \odot r \cdot a \begin{cases} = \text{CF}(\{\varphi'(x)\}, r \cdot \varepsilon) & \text{if } a \leq -d, \\ \neq \text{CF}(\{\varphi'(x)\}, r \cdot \varepsilon) & \text{if } a > -d. \end{cases}$$

Thus $\text{dist}(\varphi'(x), \varphi'(y)) = r \cdot d$. Also, for any $z \in \Gamma \setminus \Gamma_\infty$, let P be a path from x to z . If necessary, by subdividing P , we see that P is mapped by φ' to a path from $\varphi'(x)$ to $\varphi'(z)$ of length equal to r times the length of P . Since φ' is injective, distinct paths from x to z are mapped by φ' to distinct paths from $\varphi'(x)$ to $\varphi'(z)$. Since φ' is surjective, all paths from $\varphi'(x)$ to $\varphi'(z)$ are images by φ' of paths from x to z .

By Lemma 3.8(2), φ' induces a bijection between Γ_∞ and Γ'_∞ . Also, by Lemma 3.8(2) and its proof, when $\Gamma_\infty \ni x \mapsto x' \in \Gamma'_\infty$, the point x' is a (unique) accumulation point of the image $\{\varphi'(y_i)\}$ of any sequence $\{y_i\}$ such that $y_i \rightarrow x$ as $i \rightarrow \infty$. This means that φ' is continuous. By the same argument for ψ^{-1} , we obtain the desired $\frac{1}{r}$ -expansive map $\varphi := (\varphi')^{-1} : \Gamma' \rightarrow \Gamma; x' \mapsto x$. $\square$

Theorem 1.1 induces several corollaries.

Corollary 3.9. Let ψ and φ be as in Theorem 1.1. Then

$$\psi(f) = (r \cdot f) \circ \varphi$$

holds for any $f \in \text{Rat}(\Gamma)$.

We need the following lemma to prove Corollary 3.9.

Lemma 3.10. *For any $x \in \Gamma \setminus \Gamma_\infty$ and $l \in \mathbf{R}_{>0} \cup \{\infty\}$,*

$$\psi(\text{CF}(\{x\}, l)) = \text{CF}(\{\varphi^{-1}(x)\}, r \cdot l)$$

holds.

Proof. Assume that there exists $l \in \mathbf{R}_{>0} \cup \{\infty\}$ such that $\psi(\text{CF}(\{x\}, l)) \neq \text{CF}(\{\varphi^{-1}(x)\}, r \cdot l)$. Let l_1 be the infimum of such l . Since for any l such that $0 < l < l_1$,

$$\begin{aligned} \text{CF}(\{\varphi^{-1}(x)\}, r \cdot l) &= \psi(\text{CF}(\{x\}, l)) \\ &= \psi(\text{CF}(\{x\}, l_1) \oplus (-l)) \\ &= \psi(\text{CF}(\{x\}, l_1)) \oplus (-r \cdot l) \end{aligned}$$

and the minimum value of $\psi(\text{CF}(\{x\}, l_1))$ is $-r \cdot l_1$ by Lemma 3.7, the rational function $\psi(\text{CF}(\{x\}, l_1))$ coincides with $\text{CF}(\{\varphi^{-1}(x)\}, r \cdot l_1)$. Thus l_1 cannot be ∞ . Note that for any l_2 such that $l_2 > l_1$, $\psi(\text{CF}(\{x\}, l_2)) \neq \text{CF}(\{\varphi^{-1}(x)\}, r \cdot l_2)$ holds. As $\psi(\text{CF}(\{x\}, l_1)) = \psi(\text{CF}(\{x\}, l_2) \oplus (-l_1)) = \psi(\text{CF}(\{x\}, l_2)) \oplus (-r \cdot l_1)$ and the minimum value of $\psi(\text{CF}(\{x\}, l_2))$ is $-r \cdot l_2$ by Lemma 3.7, there exists a pole $y' \in \Gamma'$ of $\psi(\text{CF}(\{x\}, l_2))$ such that $\psi(\text{CF}(\{x\}, l_1))(y') = -r \cdot l_1$. Thus $\text{dist}(\varphi^{-1}(x), y') \geq r \cdot l_1$. Let y be the finite point $\varphi(y')$ of Γ . By Lemma 3.8(1), there exists $\varepsilon > 0$ for y such that $\psi(\text{CF}(\{y\}, \varepsilon)) = \text{CF}(\{y'\}, r \cdot \varepsilon)$. Then, if necessary, by replacing ε with a smaller positive number, we can assume that there exists $m' \in \mathbf{Z}_{>0}$ such that

$$\begin{aligned} &\psi(\text{CF}(\{x\}, l_1 + \varepsilon)) \\ &= \psi(\text{CF}(\{x\}, l_1 + \varepsilon)) \oplus \text{CF}(\{y'\}, r \cdot \varepsilon)^{\odot m'} \odot (-r \cdot l_1). \end{aligned}$$

Let m'_1 be the minimum integer of such m' . Then we have

$$\text{CF}(\{x\}, l_1 + \varepsilon) = \text{CF}(\{x\}, l_1 + \varepsilon) \oplus \text{CF}(\{y\}, \varepsilon)^{\odot m'_1} \odot (-l_1).$$

When $\text{dist}(x, y) > l_1$, it does not occur since $\text{CF}(\{x\}, l_1 + \varepsilon)(y) < -l_1$ and $(\text{CF}(\{y\}, \varepsilon)^{\odot m'_1} \odot (-l_1))(y) = -l_1$. Thus $\text{dist}(x, y) = l_1$. On the other hand, in this case, by the definition of chip-firing moves, we have

$$\text{CF}(\{x\}, l_1 + \varepsilon) = \text{CF}(\{x\}, l_1 + \varepsilon) \oplus \text{CF}(\{y\}, \varepsilon) \odot (-l_1).$$

Hence, $m'_1 = 1$ for any such y' . However, this means that

$$\psi(\text{CF}(\{x\}, l_1 + \varepsilon)) = \text{CF}(\{\varphi^{-1}(x)\}, r \cdot l_1 + r \cdot \varepsilon)$$

holds. It contradicts the definition of l_1 . $\square$

In the proof of Corollary 3.9, we will also use the proof of the main theorem of [12] “for a tropical curve Γ , the semifield $\text{Rat}(\Gamma)$ is a finitely generated semifield over $\mathbf{T}$ ”; in that proof, the author showed that all chip-firing moves defined by one finite point, and of the form in Lemma 3.8(2) generate $\text{Rat}(\Gamma)$ as a semifield over $\mathbf{T}$ ([12, Remark 3.7]).

Proof of Corollary 3.9. Let f (resp. g) be a chip-firing move on Γ of the form in Lemma 3.10 (resp. Lemma 3.8(2)). By Lemmas 3.8, 3.10 and the definition of φ , we have

$$\psi(f) = (r \cdot f) \circ \varphi \quad \text{and} \quad \psi(g) = (r \cdot g) \circ \varphi.$$

Since all of such f and g generate $\text{Rat}(\Gamma)$ as a semifield over $\mathbf{T}$, for any $h \in \text{Rat}(\Gamma)$, we have $\psi(h) = (r \cdot h) \circ \varphi$. In fact, if $\psi(\tilde{f}) = (r \cdot \tilde{f}) \circ \varphi$ and $\psi(\tilde{g}) = (r \cdot \tilde{g}) \circ \varphi$, then we have

$$\begin{aligned} \psi(\tilde{f} \oplus \tilde{g}) &= \psi(\tilde{f}) \oplus \psi(\tilde{g}) \\ &= (r \cdot \tilde{f}) \circ \varphi \oplus (r \cdot \tilde{g}) \circ \varphi \\ &= ((r \cdot \tilde{f}) \oplus (r \cdot \tilde{g})) \circ \varphi \\ &= (r \cdot (\tilde{f} \oplus \tilde{g})) \circ \varphi \end{aligned}$$

and

$$\begin{aligned} \psi(\tilde{f} \odot \tilde{g}) &= \psi(\tilde{f}) \odot \psi(\tilde{g}) \\ &= (r \cdot \tilde{f}) \circ \varphi \odot (r \cdot \tilde{g}) \circ \varphi \\ &= ((r \cdot \tilde{f}) \odot (r \cdot \tilde{g})) \circ \varphi \\ &= (r \cdot (\tilde{f} \odot \tilde{g})) \circ \varphi. \end{aligned}$$

Hence, we have the conclusion. $\square$

It is easy to check that the converse of Theorem 1.1 holds:

Lemma 3.11. *Let Γ, Γ' be tropical curves. An r -expansive map $\varphi : \Gamma \rightarrow \Gamma'$ with $r \in \mathbf{R}_{>0}$ induces the semiring isomorphism*

$$\psi : \text{Rat}(\Gamma') \rightarrow \text{Rat}(\Gamma); \quad f' \mapsto \left(\frac{1}{r} \cdot f' \right) \circ \varphi$$

and $\psi(1) = \frac{1}{r}$.

Proof. Since φ is surjective, $\psi(f')$ is a function $\Gamma' \rightarrow \mathbf{R} \cup \{\pm\infty\}$. For $x, y \in \Gamma$, since $r \cdot \text{dist}(x, y) = \text{dist}(\varphi(x), \varphi(y))$, slopes of f' and $\psi(f')$ on intervals corresponding by φ coincide. Thus $\psi(f')$ is a rational function on Γ' .

Let

$$\psi' : \text{Rat}(\Gamma) \rightarrow \text{Rat}(\Gamma'); \quad f \mapsto (r \cdot f) \circ \varphi^{-1}.$$

By a similar argument above, we have $\psi'(f) \in \text{Rat}(\Gamma')$ for any $f \in \text{Rat}(\Gamma)$. By definition, we have

$$\psi' \circ \psi = \text{id}_{\text{Rat}(\Gamma')} \quad \text{and} \quad \psi \circ \psi' = \text{id}_{\text{Rat}(\Gamma)},$$

where $\text{id}_{\text{Rat}(\Gamma')}$ (resp. $\text{id}_{\text{Rat}(\Gamma)}$) denotes the identity map of $\text{Rat}(\Gamma')$ (resp. $\text{Rat}(\Gamma)$). Hence ψ is bijective.

For $f', g' \in \text{Rat}(\Gamma')$, we have

$$\begin{aligned} \psi(f' \oplus g') &= \left(\frac{1}{r} \cdot (f' \oplus g') \right) \circ \varphi \\ &= \left(\frac{1}{r} \cdot f' \oplus \frac{1}{r} \cdot g' \right) \circ \varphi \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{1}{r} \cdot f' \right) \circ \varphi \oplus \left(\frac{1}{r} \cdot g' \right) \circ \varphi \\
&= \psi(f') \oplus \psi(g')
\end{aligned}$$

and

$$\begin{aligned}
\psi(f' \odot g') &= \left(\frac{1}{r} \cdot (f' \odot g') \right) \circ \varphi \\
&= \left(\frac{1}{r} \cdot f' \odot \frac{1}{r} \cdot g' \right) \circ \varphi \\
&= \left(\frac{1}{r} \cdot f' \right) \circ \varphi \odot \left(\frac{1}{r} \cdot g' \right) \circ \varphi \\
&= \psi(f') \odot \psi(g').
\end{aligned}$$

Hence, ψ is a semiring isomorphism. $\square$

Now we can prove Corollary 1.2:

Proof of Corollary 1.2. Clearly, both $\mathcal{C}, \mathcal{D}$ are groupoids.

Let

$$F : \mathcal{C} \rightarrow \mathcal{D}$$

be

$$\text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D}); \quad \Gamma \mapsto \Gamma$$

and for $\Gamma, \Gamma' \in \text{Ob}(\mathcal{C})$,

$$\text{Hom}_{\mathcal{C}}(\Gamma, \Gamma') \rightarrow \text{Hom}_{\mathcal{D}}(\Gamma, \Gamma); \quad \psi \mapsto \varphi,$$

where φ is the $\frac{1}{\psi(1)}$ -expansive map $\Gamma' \rightarrow \Gamma$ defined as in Theorem 1.1. Let

$$G : \mathcal{D} \rightarrow \mathcal{C}$$

be

$$\text{Ob}(\mathcal{D}) \rightarrow \text{Ob}(\mathcal{C}); \quad \Gamma \mapsto \Gamma$$

and for $\Gamma, \Gamma' \in \text{Ob}(\mathcal{D})$,

$$\text{Hom}_{\mathcal{D}}(\Gamma, \Gamma') \rightarrow \text{Hom}_{\mathcal{C}}(\Gamma, \Gamma); \quad \varphi \mapsto \left(f' \mapsto \left(\frac{1}{r} \cdot f' \right) \circ \varphi \right),$$

where φ is an r -expansive map. By Theorem 1.1 and Lemma 3.11, F and G are defined. Then F and G are (contravariant) functors. By Corollary 3.9, we have $G \circ F = \text{id}_{\mathcal{C}}$. Clearly, $F \circ G = \text{id}_{\mathcal{D}}$ holds. $\square$

Finally, we will describe all semiring automorphisms of tropical curves. To do so, we start characterizing star-shaped tropical curves consisting of a finite number (≥ 1) of $[0, \infty]$:

Lemma 3.12. *Let Γ be a tropical curve. Then the following are equivalent:*

- (1) *there exists a model (G, l) for Γ such that $l(E(G)) = \{\infty\}$; and*
- (2) *Γ is a star-shaped tropical curve consisting of a finite number (≥ 1) of $[0, \infty]$.*

Proof. Clearly (2) implies (1).

Assume that (2) does not hold. If the genus $g(\Gamma)$ is at least one, then, clearly, (1) does not hold. Let $g(\Gamma) = 0$. If Γ is a singleton, then (1) does not hold. Assume that Γ is not a singleton. If Γ has no points of valence at least three, then there exists a unique number $a \in \mathbf{R}_{>0}$ such that Γ is isometric to $[0, a]$. Hence, in this case, (1) does not hold. Assume that Γ has some points of valence at least three. If such points are only one, then Γ is star-shaped. Since (2) does not hold, the underlying graph $G_\circ$ of the canonical model $(G_\circ, l_\circ)$ for Γ has a leaf edge of a finite length. Hence (1) does not hold. Assume that there exist at least two points x, y of valence at least three. Since there exist only a finite number of such points, we can assume that the only path P from x to y contains no such points other than x, y . Then P is an edge of $G_\circ$ and has a finite length. This means that (1) does not hold. $\square$

Corollary 3.13. *Let Γ be a tropical curve. Then the following are equivalent:*

- (1) *there exists a model (G, l) for Γ such that $l(E(G)) = \{\infty\}$; and*
- (2) *$\text{Aut}_{\mathbf{T}}(\text{Rat}(\Gamma)) \neq \text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma))$.*

Proof. If Γ is a singleton, then Γ does not satisfy either (1) or (2) by Corollary 3.4 since $\text{Rat}(\Gamma) = \mathbf{T}$.

Assume that Γ is not a singleton. We will show that if (1) does not hold, then (2) does also not hold. The group $\text{Aut}_{\mathbf{T}}(\text{Rat}(\Gamma))$ is a subgroup of $\text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma))$. For $\psi \in \text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma))$, let $r := \psi(1)$ and φ be the $\frac{1}{r}$ -expansive map $\Gamma \rightarrow \Gamma$ corresponding to ψ . Suppose that Γ is homeomorphic to a circle S^1 . Let x, y be antipodal points and P_1, P_2 be the distinct paths from x to y . Then these paths have the same length $\text{dist}(x, y)$. Hence $\varphi(P_1)$ and $\varphi(P_2)$ are distinct paths from $\varphi(x)$ to $\varphi(y)$ of length $\frac{1}{r} \cdot \text{dist}(x, y)$. Therefore we have $r = 1$ and thus $\psi \in \text{Aut}_{\mathbf{T}}(\text{Rat}(\Gamma))$. If Γ is not homeomorphic to a circle S^1 , then Γ has at least one point of valence other than two. For the canonical model $(G_\circ, l_\circ)$ for Γ , there exists an edge $e \in E(G_\circ)$ such that $l_\circ(e)$ is finite since (1) does not hold. By Lemma 3.12, the endpoints v, w of e (possibly $v = w$) have valences other than two. Since φ is a homeomorphism, it respects valences, and thus $\varphi(e) \in E(G_\circ)$. If $r < 1$, then $l_\circ(e) < l_\circ(\varphi(e))$. By the same argument, we have the following sequences:

$$e, \varphi(e), \varphi^2(e), \dots \in E(G_\circ)$$

and

$$l_\circ(e) < l_\circ(\varphi(e)) < l_\circ(\varphi^2(e)) < \dots$$

These sequences mean that $\#E(G_\circ) = \infty$. It is a contradiction. When $r > 1$, we have a contradiction by a similar argument. Hence r must be one, and thus (2) does not hold.

Assume that (1) holds. Then by Lemma 3.12, Γ is a star-shaped tropical curve consisting of a finite number (≥ 1) of $[0, \infty]$. Clearly, for any $r \in \mathbf{R}_{>0}$, there exists an r -expansive map $\Gamma \rightarrow \Gamma$ that is a dilatation. By Lemma 3.11, this induces a semiring automorphism $\psi : \text{Rat}(\Gamma) \rightarrow \text{Rat}(\Gamma)$ such that $\psi(1) = \frac{1}{r}$. Thus $\text{Aut}_{\mathbf{T}}(\text{Rat}(\Gamma))$ is a proper subgroup of $\text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma))$. $\square$

Proposition 3.14. *Let Γ be a star-shaped tropical curve consisting of a finite number $n \geq 1$ of $[0, \infty]$.*

- (1) *If $n \neq 2$, then*

$$\text{Aut}(\Gamma) \cong S_n$$

and

$$\text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma)) \cong \text{Aut}(\Gamma) \times \mathbf{R}_{>0},$$

where S_n denotes the symmetric group of degree n and $\mathbf{R}_{>0}$ is regarded as the group $\mathbf{R}_{>0}$ with the usual multiplication as its binary operation.

(2) Let $n = 2$ and Γ be identified with $[-\infty, \infty]$ and ι be the inversion with respect to 0. Then

$$\text{Aut}(\Gamma) = \langle \{ \text{the inversion with respect to } x \mid x \in \Gamma \setminus \Gamma_\infty \} \rangle \cong \mathbf{R} \rtimes \langle \iota \rangle$$

and

$$\text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma)) \cong \text{Aut}(\Gamma) \rtimes \mathbf{R}_{>0},$$

where $\langle \cdot \rangle$ denotes the group generated by $\cdot$, and $\mathbf{R}$ is regarded as the group $\mathbf{R}$ with the usual addition as its binary operation, and $\mathbf{R}_{>0}$ as the group $\mathbf{R}_{>0}$ with the usual multiplication as its binary operation.

Proof. Let R denote the set of all r -expansive maps $\Gamma \rightarrow \Gamma$ with all $r \in \mathbf{R}_{>0}$. By Corollary 1.3, this R is isomorphic to $\text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma))$ as a group.

We shall prove (1). Let $n = 1$ and $\Gamma = [0, \infty]$. For any $r > 0$ and any r -expansive map $\varphi : \Gamma \rightarrow \Gamma$, the point $\varphi(0)$ must be zero. In fact, since φ is a homeomorphism, $\varphi(0) = 0$ or ∞ . As $\text{dist}(0, 1) = 1$, we have $\text{dist}(\varphi(0), \varphi(1)) = r$. As the point $\varphi(1)$ is in $(0, \infty)$, the point $\varphi(0)$ cannot be ∞ . Thus R is isomorphic to $\mathbf{R}_{>0}$ as a group. On the other hand, $\text{Aut}(\Gamma)$ consists of only the identity of Γ . Thus $\text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma))$ is isomorphic to $\text{Aut}(\Gamma) \times \mathbf{R}_{>0}$ as a group.

Let $n \geq 3$. There exists a unique point $x \in \Gamma$ of valence n . Since any r -expansive map $\varphi : \Gamma \rightarrow \Gamma$ with any $r \in \mathbf{R}_{>0}$ is a homeomorphism, $\varphi(x) = x$ holds. Thus $\text{Aut}(\Gamma)$ is isomorphic to the symmetric group of degree n as a group. This φ defines the pair of an element of $\text{Aut}(\Gamma)$ and a dilatation of Γ with r as its expansion factor. Conversely, the pair of an element of $\text{Aut}(\Gamma)$ and a dilatation of Γ with expansion factor r for each $r \in \mathbf{R}_{>0}$ defines an r -expansive map $\Gamma \rightarrow \Gamma$. An element of $\text{Aut}(\Gamma)$ and a dilatation of Γ with expansion factor any $r \in \mathbf{R}_{>0}$ are commutative, and thus we have the conclusion.

We shall prove (2). Since $\Gamma = [-\infty, \infty]$, the first assertion is clear. We show that $\text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma)) \cong \text{Aut}(\Gamma) \rtimes \mathbf{R}_{>0}$. Let $\varphi : \Gamma \rightarrow \Gamma$ be an r -expansive map with $r \in \mathbf{R}_{>0} \setminus \{1\}$. Then φ has a unique fixed point x other than $\pm\infty$. In fact, it is easy to check that if $\varphi(\infty) = \infty$ and $\varphi(-\infty) = -\infty$, then $x = \frac{\varphi(0)}{1-r}$, and that if $\varphi(\infty) = -\infty$ and $\varphi(-\infty) = \infty$, then $x = \frac{\varphi(0)}{1+r}$. In both cases, since $r \neq 1$, other finite points are not fixed.

We consider the group consisting of all dilatation $\theta_r : \Gamma \rightarrow \Gamma$ fixing 0 with r as its expansion factor for any $r \in \mathbf{R}_{>0}$, which is isomorphic to $\mathbf{R}_{>0}$ as a group. If $\varphi(\infty) = \infty$ and $\varphi(-\infty) = -\infty$, then $\varphi = (\varphi_x \circ \text{id}_\Gamma) \circ \theta_r$, where φ_x denotes the translation $\Gamma \rightarrow \Gamma; y \mapsto y + x$. If $\varphi(\infty) = -\infty$ and $\varphi(-\infty) = \infty$, then $\varphi = (\varphi_x \circ \iota) \circ \theta_r$. Conversely, an element of $\text{Aut}(\Gamma)\mathbf{R}_{>0}$ defines an element of R . Thus $R = \text{Aut}(\Gamma)\mathbf{R}_{>0}$. Clearly both $\text{Aut}(\Gamma)$ and $\mathbf{R}_{>0}$ are subgroups of R , and for $\psi \in R$ and $\varphi \in \text{Aut}(\Gamma)$, the composition $\psi \circ \varphi \circ \psi^{-1}$ is a 1-expansive map, i.e., $\psi \circ \varphi \circ \psi^{-1} \in \text{Aut}(\Gamma)$. Hence $\text{Aut}(\Gamma)$ is a normal subgroup of R . Thus we have the conclusion. $\square$

Note that in Proposition 3.14(2), since there exist an element of $\text{Aut}(\Gamma)$ and an element of $\mathbf{R}_{>0}$ which are not commutative, we have $\text{Aut}_{\text{semiring}}(\text{Rat}(\Gamma)) \not\cong \text{Aut}(\Gamma) \times \mathbf{R}_{>0}$. For example,

$$((\varphi_1 \circ \text{id}_\Gamma) \circ \theta_2)(0) = 1$$

and

$$(\theta_2 \circ (\varphi_1 \circ \text{id}_\Gamma))(0) = 2,$$

and thus

$$(\varphi_1 \circ \text{id}_I) \circ \theta_2 \neq \theta_2 \circ (\varphi_1 \circ \text{id}_I).$$

CRediT authorship contribution statement

JuAe Song: Conceptualization, Visualization, Writing – original draft, Writing – review & editing, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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